From: **Boris Beranger** b.beranger@unsw.edu.au
Subject: Re: Interior extremes manuscript
Date: 22 September 2026 at 15:00
To: Peng Zhong pzhong@uow.edu.au
Cc: Scott Sisson scott.sisson@unsw.edu.au

Hi Peng,

Let's discuss this point on Friday. Meanwhile, do you think that we can complete the points related to the algorithm (2 & 3 from my previous email)?

Regarding Friday, when will you be around? We should make sure that we have enough time to discuss our current project(s).

Best,

Dr. Boris Béranger

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From: Peng Zhong <pzhong@uow.edu.au>
Sent: Tuesday, 15 September 2026 9:45 AM
To: Boris Beranger <b.beranger@unsw.edu.au>
Cc: Scott Sisson <scott.sisson@unsw.edu.au>
Subject: Re: Interior extremes manuscript

The area in the Florida application is actually quite small. It is basically a city wide area. Let's say we want to model the data in {X: ||X||_p > u_p}, then we can have {X : ||X||_1 > u_1}, where u_1 = u_p*D^{1-1/p} based on the holder's inequality. In the case of $p = \infty$, we have u_1 = u_p*D. This is to say for Pareto tailed X, the average should be above u_p. Since u_p is selected to be a high quantiles from the maximum, the chances of getting any data such that the average is above this high threshold u_p is almost impossible.

The acceptance probability you describe here I believe is for sampling data for risk functional Lp norm from the samples with risk functional L1 norm using rejection sampling. The acceptance ratio ranges from 1/sqrt(D) for r_2 to 1/D for p = \infty. This is the other way around.

Sorry for my late reply.

Best,
Peng

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> On 10 Sep 2026, at 23:46, Boris Beranger <b.beranger@unsw.edu.au> wrote:
>
> Hi Peng,
>
> Thanks for providing answers to those points.
>
> First point:

First point:

I understand what you are hinting at and I agree with you. I've worked out the expression of the acceptance probability from the Dombry, Legrand & Opitz paper and it turns out, assuming unit-Pareto margins, it can be written as $D^{1/p-1} \nu(\{ \bm x: r_0(\bm x) > 1\}) / \nu(\{ \bm x: r_p(\bm x) > 1\})$ where you can recognize the norming constants for the distributions of the r_0 and r_p Pareto processes. See my notes for some derivations.

What this means is that the acceptance ratio depends on the dependence structure. On the last page of my notes you can see some bounds for the acceptance probability raging from $D^{1/p-1}$ when we have asymptomatic independence, to 1 when we have complete dependence. All this makes sense to me.

This brings me to the following comments:

1. In the Florida data $D=3600$ and because the study region is big, the overall level of dependence is probably quite weak. As such the acceptance probability is small and, as you mentioned you only get a very small number of observations. By the way, how many was it exactly, 9? I suggest we try fitting the model, using the same inference techniques but to sub-region. The third comment from referee 2 suggest to use a smaller spatial domain so that might be useful here. Could you try to fit the model to smaller regions and see if our performance is still good? Maybe try dimensions in the order of 200-300?
2. We could compare the acceptance probability from the Dombry, Legrand & Opitz rejection sampling algorithm that are given in Table 1 to the true one and we should have a match.
3. Can we derive the acceptance probability from your improved algorithm in the similar fashion as what I did from the reference algorithm? This would make the paper more solid.
4. Could we have an inference method based on the improved rejection-sampling algorithm? This would give us more samples.

I don't think any of the above would be too much work but they would strengthen the paper and allow us to keep using other norms as risk functions.

As I said previously, we can discuss online if that makes things easier.

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From: Peng Zhong <pzhong@uow.edu.au>
Sent: Thursday, 10 September 2026 2:21 PM
To: Boris Beranger <b.beranger@unsw.edu.au>
Cc: Scott Sisson <scott.sisson@unsw.edu.au>
Subject: Re: Interior extremes manuscript

Hi Boris,

The thing is that: If we simulate X from r-Pareto with r1, then we can use r0 with a higher threshold to produce the data from r-Pareto with r0. This is what we did in the simulation. However, this procedure requires two steps of selection, and in applications, it will produce almost no observations for fitting as the higher threshold scales up quickly when dimension is high. Let's say r1 = max, and r0=average, we know that r0 < r1. If we have samples X from r1 with threshold u, that is r1(X)>u, to produce samples from r0, we can simply do r0(X) > u. However, u is just too large to pick up any samples, i.e., {X: r0(X) > u} is empty. Therefore what we did in the first place is wrong and not practical.

For the second point, when only the risk function is different, and the data is regular varying and in the maximum domain of attraction of a max-stable processes with intensity function $lambda$, then, for different risk functional, the only thing changes will be the normalising constant that makes the intensity function a proper density. However, real data does not follow the exact distribution. Selecting data using different risk functional leads to different parameter estimates. This can be seen from the Theorem 3 in the Dombry's Paper, as the same measure σ in equation 6 are used to compute the σ_ℓ.

These are the points I want to make in the revision. Sorry for the confusion.
Happy to have a zoom meeting to discuss.

Best,
Peng

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On 9 Sep 2026, at 14:33, Boris Beranger <b.beranger@unsw.edu.au> wrote:

Dombry & Ribatet (2015) paper attached

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From: Boris Beranger
Sent: Wednesday, 9 September 2026 2:29 PM
To: Peng Zhong <pzhong@uow.edu.au>
Cc: Scott Sisson <scott.sisson@unsw.edu.au>
Subject: Interior extremes manuscript

Hi Peng,

I'm going through the interior extremes paper that you have revised and have a couple of questions for you:

- For some reason (that are not 100% clear to me, even after reading the new paragraph on p9) we now only focus on r-Pareto processes with L1 norm risk functions. What I wonder is how come was everything previously working as expected in the simulations and real data analysis but now we have a reason for not doing it? It feels that we are restricting the contributions of the paper.
- In that same revised paragraph you mention that no matter the risk function, the r-Pareto process have the same spectral measure. I think they share the same underlying exponent measure coming from the max-stable process but the spectral (probability) measure of the r-Pareto model itself does depend on r. In fact Dombry & Ribatet (2015, attached) show in Proposition 2 the link between the spectral measures when using two different risk functions. Furthermore, Proposition 4 gives a non-parametric estimator of the spectral measure for a risk function, say r_1, when the thresholding has been done with the risk function r_2. I think a parametric version of this estimator is not too difficult to establish.
  My question here (and overall) is: can we find a way to justify what we previously did for other risk functionals than L1?

Happy to discuss online if needed

Best,

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<SII.2015.v8.n1.a2.pdf>

<r-Pareto-RejectionSampling.pdf>